

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

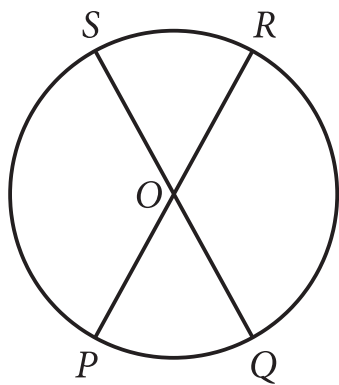
Circles

10 questions · ~15 min

01

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- The center of the circle is point O .
- Points S , R , Q , and P are on the circle.
- Line segment PR is a diameter of the circle.
- Line segment QS is a diameter of the circle.
- Diameters PR and QS intersect at point O .
- A note indicates the figure is not drawn to scale.

The circle shown has center O , circumference 144π , and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ . What is the length of arc QR ?

- A. 24π
- B. 48π
- C. 72π
- D. 96π

The measure of angle Z is 60° . What is the measure, in radians, of angle Z ?

A. $\frac{1}{6}\pi$

B. $\frac{1}{3}\pi$

C. $\frac{2}{3}\pi$

D. 1π

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

What is the radius of the circle in the xy -plane defined by $x + 2^2 + y + 5^2 = 169$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

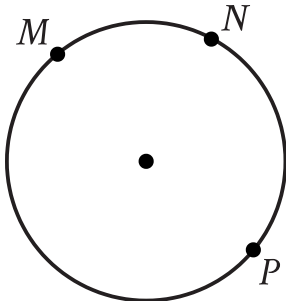
GEOMETRY AND TRIGONOMETRY

An angle has a measure of $\frac{9\pi}{20}$ radians. What is the measure of the angle in degrees?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Clockwise from top left, the following points are on the circle:
 - upper M
 - upper N
 - upper P

Points M , N , and P lie on the circle shown. On this circle, minor arc MN has a length of 39 centimeters and major arc MPN has a length of 195 centimeters. What is the circumference, in centimeters, of the circle shown?

- A. 39
- B. 156
- C. 195
- D. 234

A circle in the xy -plane has its center at $-4, 5$ and the point $-8, 8$ lies on the circle. Which equation represents this circle?

A. $x - 4^2 + y + 5^2 = 5$

B. $x + 4^2 + y - 5^2 = 5$

C. $x - 4^2 + y + 5^2 = 25$

D. $x + 4^2 + y - 5^2 = 25$

In the xy -plane, a circle with radius 5 has center $(-8, 6)$. Which of the following is an equation of the circle?

A. $(x - 8)^2 + (y + 6)^2 = 25$

B. $(x + 8)^2 + (y - 6)^2 = 25$

C. $(x - 8)^2 + (y + 6)^2 = 5$

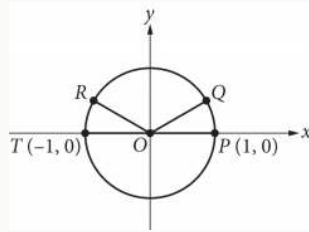
D. $(x + 8)^2 + (y - 6)^2 = 5$

GRID-IN GEOMETRY AND TRIGONOMETRY

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

CIRCLES — SET 1 · GEOMETRY AND TRIGONOMETRY



In the xy -plane above, points P, Q, R, and T lie on the circle with center O. The degree measures of angles POQ and ROT are each 30° . What is the radian measure of angle QOR ?

- A. $\frac{5}{6} \pi$
- B. $\frac{3}{4} \pi$
- C. $\frac{2}{3} \pi$
- D. $\frac{1}{3} \pi$

GRID-IN

An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in degrees?

.	/	.	/	.	.
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

CIRCLES — SET 1 · GEOMETRY AND TRIGONOMETRY